

**Probabilistic reasoning** = Knowledge + Evidence → Update Belief about the unknown.

A **joint probability distribution** gives the probability of every possible combination of random variables.

A **probabilistic model** is a joint probability distribution over a set of random variables.

Joint distribution stores complete cases;

**event** probability is obtained by summing all matching complete cases.

**Marginal Distribution**: Probability of one variable after summing over the others.

**Conditional Probability**: The probability of an event given another event.

**Product Rule** Conditional × Given = Joint  $P(A, B) = P(A | B)P(B)$

The **Chain Rule** expresses a joint probability as the product of conditional probabilities.

$$P(A, B, C) = P(A | B, C)P(B | C)P(C)$$

**Bayes' rule** computes the probability of a cause given observed evidence using the reverse conditional probability.

A **Bayesian Network** represents a large joint distribution by decomposing it into small local conditional probabilities.

A **probabilistic model** is a simplified representation of the real world using probability distributions. It is used to perform probabilistic reasoning under uncertainty.

Bayesian Network = DAG + CPT = Structure + Probabilities.

**Inference by Enumeration**:

1. Filter: keep only entries consistent with evidence e
2. Sum out hidden variables H
3. Normalize to get  $P(X|e)$

**Prior Sampling** generates complete samples by sampling each variable in topological order according to its conditional probability.

**Rejection Sampling** Generate everything, then throw away samples inconsistent with the evidence.

**Likelihood Weighting** Fix the evidence, sample the remaining variables, and assign each sample a weight.

+It uses the evidence during sampling and avoids rejecting many samples.

-Evidence affects downstream variables, but not upstream variables.

**Gibbs Sampling** repeatedly resamples one non-evidence variable while keeping evidence fixed.

+Evidence influences both upstream and downstream variables. Usually more efficient than Likelihood Weighting.

-Needs many iterations to converge.

**Independence** means that knowing one variable does not change the probability of the other.

**Conditional independence** means that once Z is known, knowing Y does not change the probability of X.

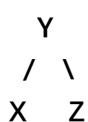
**D-separation** is a graphical method for determining whether two variables are conditionally independent given some evidence.

**Causal Chains**  $X \rightarrow Y \rightarrow Z$

No evidence → Dependent

Observe Y → Independent

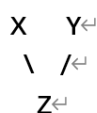
**Common Cause**



No evidence → Dependent

Observe Y → Independent

**Common Effect**



No evidence → Independent

Observe Y → Dependent

A **Decision Network** extends a Bayesian Network by adding decision nodes and utility nodes. The goal is to choose the action that maximizes the expected utility (MEU).

**MEU (Maximize Expected Utility)** Choose the action with the highest expected utility.

**Utility** How good or bad an outcome is

**VPI (Value of Perfect Information)** = The increase in maximum expected utility after obtaining perfect information.

$VPI = MEU(\text{after information}) - MEU(\text{before information})$